

Solution of Exercise 3

Exercise 3.1 Sequential quadratic programming

a) Lagrangian function:

$$\begin{aligned} L(\mathbf{x}, \boldsymbol{\mu}) &= f(\mathbf{x}) + \boldsymbol{\mu}^T \mathbf{h}(\mathbf{x}) \\ &= 5x_1^5 + 25x_1x_2 + 10x_2^3 + \mu_1(x_1x_2^3 - 2) + \mu_2(-x_1 - 2x_1x_2 + 2) + \mu_3(-x_1) \end{aligned}$$

Partial derivatives of the Lagrangian at iteration point $\mathbf{x}^k, \boldsymbol{\mu}^k$:

$$\nabla_{\mathbf{x}} L(\mathbf{x}, \boldsymbol{\mu}) = \begin{bmatrix} 25x_1^4 + 25x_2 + x_2^3\mu_1 + (-1 - 2x_2)\mu_2 - \mu_3 \\ 25x_1 + 30x_2^2 + 3x_1x_2^2\mu_1 - 2x_1\mu_2 \end{bmatrix}$$

$$\nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^k, \boldsymbol{\mu}^k) = \begin{bmatrix} 100x_1^3 & 25 + 3x_2^2\mu_1 - 2\mu_2 \\ 25 + 3x_2^2\mu_1 - 2\mu_2 & 60x_2 + 6x_1x_2\mu_1 \end{bmatrix}_{\mathbf{x}=\mathbf{x}^k, \boldsymbol{\mu}=\boldsymbol{\mu}^k} = \begin{bmatrix} 100 & 25 \\ 25 & 60 \end{bmatrix}$$

Gradient of f :

$$\nabla f(\mathbf{x}^k) = \begin{bmatrix} 25x_1^4 + 25x_2 \\ 25x_1 + 30x_2^2 \end{bmatrix}_{\mathbf{x}=\mathbf{x}^k} = \begin{bmatrix} 50 \\ 55 \end{bmatrix}$$

Gradients of the constraints:

$$\nabla h_1(\mathbf{x}^k) = \begin{bmatrix} x_2^3 \\ 3x_1x_2^2 \end{bmatrix}_{\mathbf{x}=\mathbf{x}^k} = \begin{bmatrix} 1 \\ 3 \end{bmatrix},$$

$$h_1(\mathbf{x}^k) = 1 - 2 = -1$$

$$\nabla h_2(\mathbf{x}^k) = \begin{bmatrix} -1 - 2x_2 \\ -2x_1 \end{bmatrix}_{\mathbf{x}=\mathbf{x}^k} = \begin{bmatrix} -3 \\ -2 \end{bmatrix},$$

$$h_2(\mathbf{x}^k) = -1 - 2 + 2 = -1$$

$$\nabla h_3(\mathbf{x}^k) = \begin{bmatrix} -1 \\ 0 \end{bmatrix},$$

$$h_3(\mathbf{x}^k) = -1$$

$\Rightarrow$ LQ approximation:

$$\begin{aligned} \min_{\mathbf{s} \in \mathbb{R}^2} \quad & \frac{1}{2} \mathbf{s}^T \begin{bmatrix} 100 & 25 \\ 25 & 60 \end{bmatrix} \mathbf{s} + \begin{bmatrix} 50 \\ 55 \end{bmatrix}^T \mathbf{s} \\ \text{s.t.} \quad & s_1 + 3s_2 - 1 \leq 0 \\ & -3s_1 - 2s_2 - 1 \leq 0 \\ & -s_1 - 1 \leq 0 \end{aligned}$$

b) Add-On: Graphical solution of the approximation

Constraints:

$$\begin{aligned} s_2 &\leq \frac{1}{3} - \frac{1}{3}s_1 \\ s_2 &\geq -\frac{1}{2} - \frac{3}{2}s_1 \\ s_1 &\geq -1 \end{aligned}$$

Optimal solution of the LQ approximation: $\mathbf{s}^* = \mathbf{s}^k = [0, -0.5]^T$

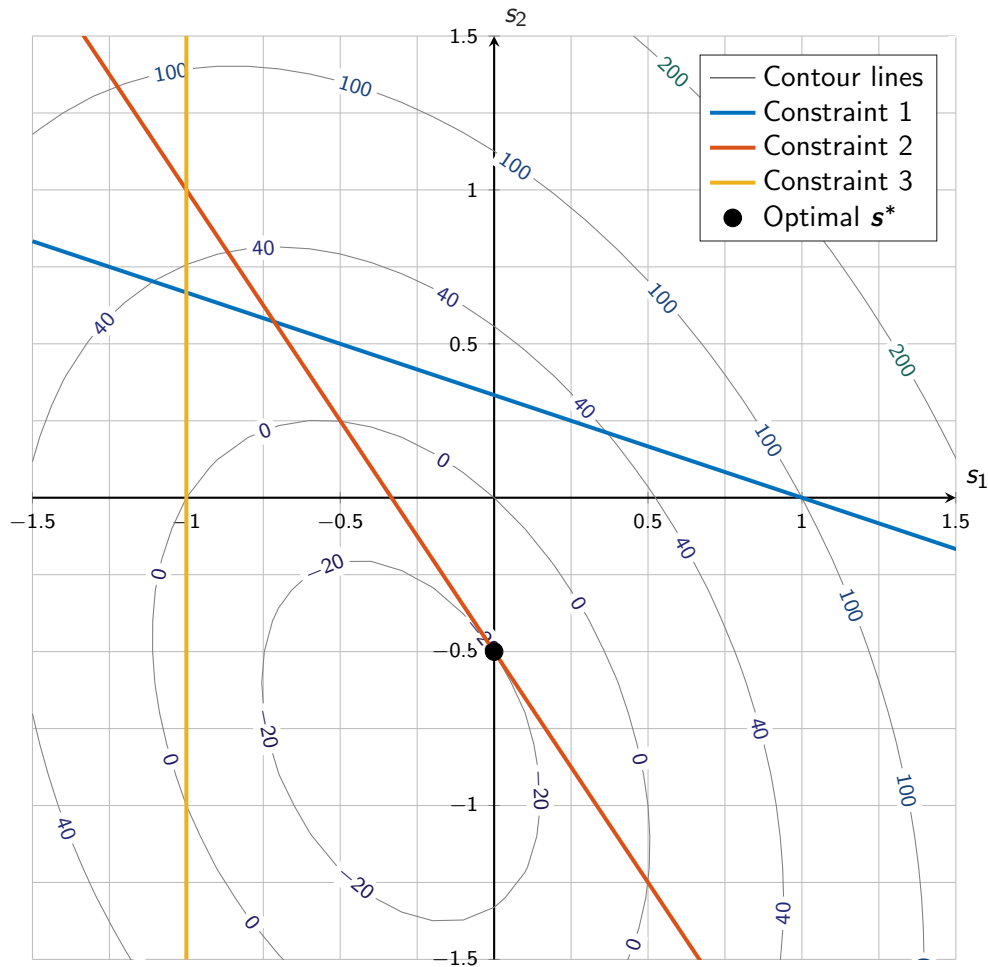


Figure 1: Graphical solution of the linear quadratic approximation.

c) Calculation of the new iteration point with full step size $\alpha = 1$:

$$\mathbf{x}^{k+1} = \mathbf{x}^k + \alpha^k \mathbf{s}^k \stackrel{\alpha=1}{=} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \end{bmatrix}$$

Check nonlinear constraints at new iteration point (i.e. check if the full step size $\alpha = 1$ is allowed, or if one of the nonlinear constraints is violated at the new iteration point):

$$h_1(\mathbf{x}^{k+1}) = 1\left(\frac{1}{2}\right)^3 - 2 = \frac{1}{8} - \frac{16}{8} = -\frac{15}{8} \leq 0$$

$$h_2(\mathbf{x}^{k+1}) = -1 - 2\frac{1}{2} + 2 = 0 \leq 0$$

$$h_3(\mathbf{x}^{k+1}) = -1 \leq 0$$

$\Rightarrow$ Full step is possible.

The nonlinear constraints can directly be considered in the calculation of the step size α^k by using a penalty function, i.e. the l_1 -penalty function

$$\begin{aligned} P(\mathbf{x}, \sigma) &= f(\mathbf{x}) + \sigma \left(\sum_{i=1}^p |g_i(\mathbf{x})| + \sum_{i=1}^q \max\{0, h_i(\mathbf{x})\} \right) \\ &= 5x_1^5 + 25x_1x_2 + 10x_2^3 \\ &\quad + \sigma \left(\max\{0, x_1x_2^3 - 2\} + \max\{0, -x_1 - 2x_1x_2 + 2\} + \max\{0, -x_1\} \right) \end{aligned}$$

Then the step size is determined with a line search method to satisfy $P(\mathbf{x}^k + \alpha^k \mathbf{s}^k, \sigma) < P(\mathbf{x}^k, \sigma)$.

Exercise 3.2 Exact penalty function

a) Set up Lagrangian function:

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(x - 1).$$

From this follow the KKT conditions

$$\begin{aligned}\nabla_x \mathcal{L}(x, \lambda) &= 2x + \lambda = 0 \\ x - 1 &= 0\end{aligned}$$

$$\Rightarrow x^* = 1 \text{ and } \lambda^* = -2.$$

b) Differences between the penalty functions:

- l_2 -penalty function:

$$\Psi_{l_2}(x) = x^2 + \sigma(x - 1)^2$$

Calculate minimum:

$$\begin{aligned}\frac{\partial \Psi_{l_2}}{\partial x} &= 2x + 2\sigma(x - 1) \\ &= 2(1 + \sigma)x - 2\sigma \stackrel{!}{=} 0 \\ \Rightarrow x &= \frac{2\sigma}{2(1 + \sigma)} = \frac{\sigma}{1 + \sigma} \xrightarrow{\sigma \rightarrow \infty} 1.\end{aligned}$$

Meaning that the minimum of the l_2 penalty function equals the optimal point of the original constrained problem only for the limiting case $\sigma \rightarrow \infty$.

- l_1 -penalty function:

$$\Psi_{l_1}(x) = x^2 + \sigma|x - 1|$$

Determine for which values of σ the penalty function $\Psi_{l_1}(x)$ has a unique minimum at $x = 1$, i.e. $\Psi_{l_1}(x) > \Psi_{l_1}(1)$ has to hold. Cases:

(I) : $x > 1$:

$$\Psi_{l_1}(x) = \underbrace{x^2}_{>1} + \sigma \underbrace{(x - 1)}_{>0} > \Psi_{l_1}(1) = 1 \quad \forall \sigma \geq 0$$

(II) : $x < 1$:

$$\begin{aligned}\Psi_{l_1}(x) &= x^2 + \sigma(1 - x) \\ \frac{\partial \Psi_{l_1}}{\partial x} &= 2x - \sigma < 2 - \sigma < 0 \quad \forall \sigma > 2 = |\lambda^*|.\end{aligned}$$

Meaning if $\sigma > |\lambda^*|$ is chosen the l_1 -penalty function has a negative derivative left of the point and thus a unique minimum at the optimal point of the original constrained problem. In this case one talks about an exact penalty function.

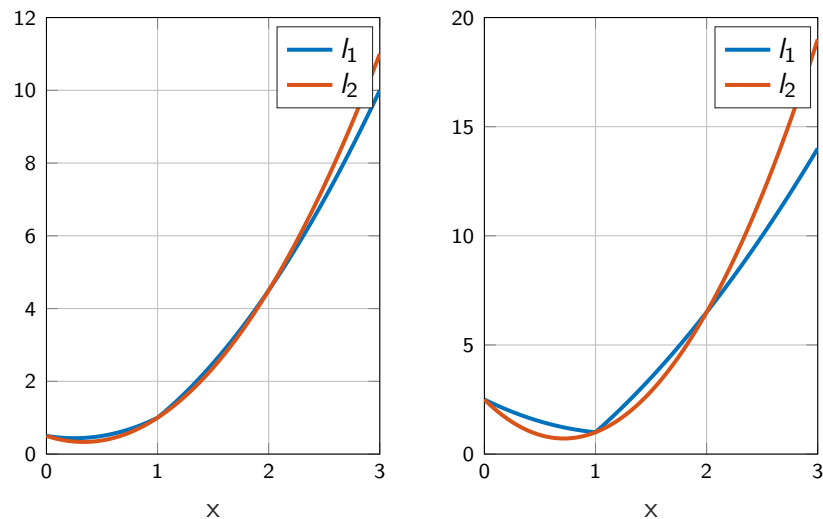


Figure 2: Curves of the penalty function for $\sigma = 0.5$ (left) and $\sigma = 2.5$ (right).

Exercise 3.3 Lagrangian duality

a) Lagrangian function

$$\begin{aligned} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{A}\mathbf{x} - \mathbf{b}) + \boldsymbol{\mu}^\top (-\mathbf{x}) \\ &= (\mathbf{A}^\top \boldsymbol{\lambda} - \boldsymbol{\mu} + \mathbf{c})^\top \mathbf{x} - \boldsymbol{\lambda}^\top \mathbf{b} \end{aligned}$$

The dual problem is

$$\max_{\boldsymbol{\lambda}, \boldsymbol{\mu}} \theta(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \begin{cases} -\boldsymbol{\lambda}^\top \mathbf{b} & \text{if } \mathbf{A}^\top \boldsymbol{\lambda} - \boldsymbol{\mu} + \mathbf{c} = \mathbf{0} \\ -\infty & \text{else,} \end{cases} \quad (2a)$$

$$\text{s.t. } \boldsymbol{\mu} \geq \mathbf{0}. \quad (2b)$$

or inserted with the corresponding values

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^2} -\boldsymbol{\lambda}^\top \mathbf{b}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} \quad (3a)$$

$$\text{s.t. } \mathbf{A}^\top \boldsymbol{\lambda} + \mathbf{c} \geq \mathbf{0}, \quad \mathbf{A}^\top = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & -3 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}. \quad (3b)$$

Here the dual variable $\boldsymbol{\lambda}$ is $\in \mathbb{R}^2$ and there are 4 inequality constraints $\rightarrow$ The dual problem is graphically solvable unlike the primal problem.

b) Add-On: Graphical solution of the dual problem: $\lambda^* = \begin{bmatrix} -1.75 \\ 0.75 \end{bmatrix}$

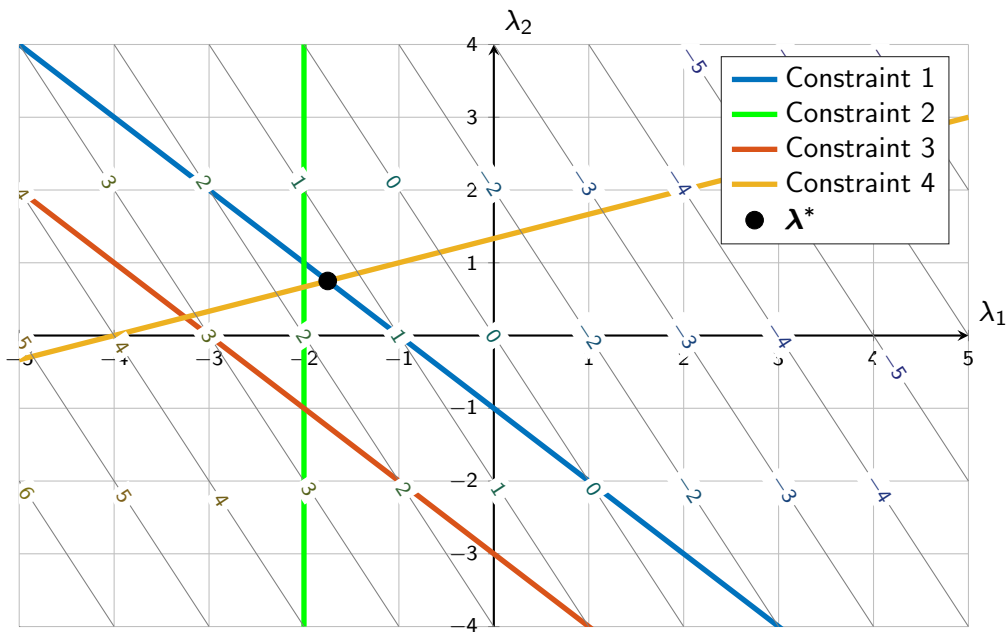


Figure 3: Graphical solution of the dual problem.

c) Calculation of the primal solution using the dual solution:

- Calculation of μ^*

$$\begin{aligned} A^T \lambda^* - \mu^* + c &= 0 \quad \Rightarrow \quad \mu^* = A^T \lambda^* + c \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} -1.75 \\ 0.75 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.25 \\ 2 \\ 0 \end{bmatrix} \end{aligned}$$

- Since $\mu_{2,3}^* \neq 0$ and due to the complementarity condition ($\mu_i^* x_i^* = 0, \quad i = 1, \dots, 4$) for the optimal primal solution $x_{2,3}^* = 0$ must hold. The remaining solution can be determined via the two equality constraints $Ax^* = b$ of the primal problem.

$$\begin{aligned} x_1^* + x_4^* &= 1 \\ x_1^* - 3x_4^* &= \frac{1}{2} \end{aligned}$$

$$\Rightarrow x^* = [0.875, 0, 0, 0.125]^T$$

- Duality gap: $c^T x^* = -\lambda^{*T} b = 1.375 \quad \Rightarrow \quad \text{No duality gap.}$